

EDUCATION

- **Purdue University, West Lafayette** 2023 - Present
Ph.D. in Mechanical Engineering GPA: 3.70/4.0
- **University of Michigan, Ann Arbor** 2020 - 2022
M.S. in Robotics GPA: 3.92/4.0
- **Purdue University, West Lafayette** 2014 - 2020
B.S.E. in Mechanical Engineering GPA: 3.75/4.0

PROFESSIONAL EXPERIENCE

- **Graduate Teaching Assistant**, Purdue University, West Lafayette 2025 - Present
- **Graduate Research Assistant**, Purdue University, West Lafayette 2023 - Present
- **Research Associate**, University of Michigan, Ann Arbor 2022 - 2023
- **Graduate Research Assistant**, University of Michigan, Ann Arbor 2021 - 2022
- **Undergraduate Research Assistant**, Purdue University, West Lafayette 2019 - 2020
- **Graduate Teaching Assistant**, Purdue University, West Lafayette 2025 - Present

JOURNAL PUBLICATIONS

‡: Equal contribution

*: Corresponding author

- **Lee, H.**, Sim, Y., Lee, H., Lee, K., Jun, M., & Lee, J.* (2026). BRIDGES: Bi-directional Robotic Integration with Digital-twin for Generalized, Efficient, and Safe Automation in Manufacturing [In preparation].
- Sim, Y., **Lee, H.**, Lee, H., Paek, S., Lee, J., Yoon, B., & Jun, M.* (2026). MMRL: A Multi-Modal Avatar based Remote Interaction and Localization Framework for Digital Twin Cross-Validation in Smart Manufacturing [In preparation].
- **Lee, H.**, Sim, Y., Yoon, B., Lee, H., Park, H., Jang, J., Kim, C., & Jun, M.* (2026). Toward provably safe AI-driven robotics autonomy for manufacturing processes: A review of methodologies and future directions. *Journal of Manufacturing Systems* [Under review].
- Sohn, J., **Lee, H.**, Lee, D., Torres, A., Neira-García, J., Zhou, Q., Abdollahi, M., Rastgaar, M., & Jun, M.* (2026). HANDS: Unsupervised Segmentation for Interpretable Identification of Human Motion Primitives in Manufacturing Task. *Journal of Manufacturing Systems* [Under review].
- Jun, M.*, Sim, Y., Kim, E., **Lee, H.** (2026). Open-world condition recognition for discrete manufacturing. *CIRP Journal of Manufacturing Science and Technology* [Under review].
- Nam, J., **Lee, H.**, Han, C., Athinarayanan, R., Jun, M., & J, Lee.* (2025). C³AM: Conversational CAD/CAM Automation via Multimodal Large Language Model Agents. *Journal of Manufacturing Systems* [Under review].
- **Lee, H.**, Sim, Y., Lee, H., Yoon, B., Kim, C., & Jun, M.* (2025). PRISM: Progressive Refinement via Informed Sampling for Multi-belief Object Pose Validation. *Robotics and Computer-Integrated Manufacturing* [Under review].
- Sim, Y., Kim, E.*, **Lee, H.**, Sohn, J., & Jun, M. (2025). Structured Sound Data Augmentation Based on Spectral Analysis for Machine Tool Operational State Monitoring. *The International Journal of Advanced Manufacturing Technology*, 141, 1705-1719.

- **Lee, H.**, Han, C., Gabor, T., Sim, Y., Akin, S., Jun, M., & Lee, J.* (2025). Manufacturable Physically Unclonable Identifiers via Cold Spray: A Full-Stack Framework with Implicit Neural Representation for Scalable Product Authentication. *Journal of Intelligent Manufacturing* [Under review].
- Kim, E., **Lee, H.**, Sim, Y., Lee, J., & Jun, M.* (2025). Overcoming Sparse Run-to-Failure Data Challenges in Manufacturing: A Contrastive Mixer Framework for Remaining Useful Life Prediction. *CIRP Annals – Manufacturing Technology*, 74, 579-583.
- **Lee, H.**, Sim, Y., Han, C., Jun, M., & Lee, J.* (2025). GRAVITON: Generalized Resampling Algorithm for Visual Inference in Target Object Localization. *Computers in Industry* [Under revision].
- Han, C., Gabor, T., **Lee, H.**, Lee, J., Akin, S., & Jun, M.* (2025). Pulsed Cold Spray System for Physical Unclonable Function Generation. *Procedia CIRP*, 137, 176-180.
- Jeon, J., Sim, Y., **Lee, H.**, Han, C., Yun, D., Kim, E., Nagendra, S. L., Jun, M., Kim, Y., Lee, S., & Lee, J.* (2025). ChatCNC: Conversational Machine Monitoring via Large Language Model and Real-Time Data Retrieval Augmented Generation. *Journal of Manufacturing Systems*, 79, 504-514.
- **Lee, H.***, Toner, T., Tilbury, D., & Barton, K. (2025). GraspMixer: Hybrid of Contact Surface Sampling and Grasp Feature Mixing for Grasp Synthesis. *IEEE Transactions on Automation Science and Engineering*, 22, 10920-10936.
- Lee, J., Akin, S.* , Sim, Y., **Lee, H.**, Kim, E., Nam, J., Song, K., & Jun, M.* (2024). A stethoscope-guided interpretable deep learning framework for powder flow diagnosis in cold spray additive manufacturing. *Manufacturing Letters*, 41, 1515-1525.
- Han, C., Lee, J., **Lee, H.**, Sim, Y., Jeon, J., & Jun, M.* (2024). Zero-shot Autonomous Robot Manipulation via Natural Language. *Manufacturing Letters*, 42, 16-20.
- **Lee, H.***, Toner, T., Tilbury, D., & Barton, K. (2022). Multi-sensor aided deep pose tracking. *IFAC-PapersOnLine*, 55(37), 326-332.

CONFERENCE PROCEEDINGS & PRESENTATIONS

‡: Presenter

*: Corresponding author

- Lee, H. ‡, Sim, Y., **Lee, H.**, Yoon, B., Lee, J., Garcia, J. E. N., Rastgaar, M., Jang, J., & Jun, M.* (2026, November). High-Precision 3D Trajectory Estimation Using Wire Encoder based Module with String Dynamic Model Compensation. In 2026 International Mechanical Engineering Congress and Exposition (IMECE) [In preparation]. American Society of Mechanical Engineers.
- Jun, M. ‡*, Sim, Y., **Lee, H.**, Kim, E., & Shakouri, A. (2026, October 11-16). *Acoustic Fingerprinting for Online Condition Monitoring in Progressive Stamping* [In preparation]. International Conference on the Technology of Plasticity 2026, Jeju Island, Republic of Korea.
- Yoon, B. ‡, **Lee, H.**, Sim, Y., & Jun, M.* (2026, September). LiDAR-Integrated Coarse-to-Fine Optimization for Geometrically Consistent Triangle Splatting from Mobile Robots [Under review]. In 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) [Under review]. IEEE.
- Jun, M. ‡*, **Lee, H.**, Sim, Y., Lee, H., Yoon, B., & Bae, D. (2026, June 28- July 3). *Leveraging a Vision Foundation Model for Adaptable and Robust Object 3D Localization and Manipulation under Real Shop-Floor Environments* [Conference Presentation]. International Conference on Precision Engineering and Sustainable Manufacturing 2026, Singapore.

- Nam, J.^{‡*}, **Lee, H.**, Han, C., Athinarayanan, R., Jun, M., & J. Lee. (2026, June). Gen-Cad/cam: Generative and Interactive Cad/cam Framework for Simulation-Informed Adaptive Maintenance. In *International Manufacturing Science and Engineering Conference* [Accepted]. American Society of Mechanical Engineers.
- Sim, Y.^{‡*}, Kim, E., Sohn, J., Barr, J., **Lee, H.**, Beach, B., Jun, M. & Shakouri, A. (2026, June). Generative Latent Space Modeling of Stamping Sound for Anomaly Detection and Predictive Maintenance. In *International Manufacturing Science and Engineering Conference* [Accepted]. American Society of Mechanical Engineers.
- **Lee, H.**^{‡*}, Sim, Y., Lee, H., Yoon, B., Kim, C., & Jun, M. (2026, January 29-30). *PRISM: Progressive Refinement via Informed Sampling for Multi-belief Object Pose Estimation* [Conference Presentation]. 2026 Institute for Control, Optimization and Network Student Research Conference, West Lafayette, IN, United States.
- Han, C.[‡], Sim, Y., Jung, H., Lee, J., **Lee, H.**, Kang, Y., Woo, S., Kim, G., Park, H., & Jun, M.* (2026, July). IMPACT: Industrial Machine Perception via Acoustic Cognitive Transformer. In *Knowledge Discovery and Data Mining 2026 (KDD)* [Under review]. ACM.
- **Lee, H.**^{‡*}, Sim, Y., Han, C., Lee, J., Bera, A., Kim, C., & Jun, M. B. G. (2025, October). EASEIR: Efficient and Adaptive Safe-set Estimation via Implicit Representation for High-dimensional Motion Planning. In *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 11661-11668). IEEE.
- Kim, E., **Lee, H.**, Sim, Y., Lee, J., & Jun, M.^{‡*} (2025, August 17-23). *Overcoming Sparse Run-to-Failure Data Challenges in Manufacturing: A Contrastive Mixer Framework for Remaining Useful Life Prediction* [Conference Presentation]. 74th CIRP Annals – Manufacturing Technology, Stockholm, Sweden.
- Sim, Y.^{‡*}, Lee, J., **Lee, H.**, Han, C., Lee, H., & Jun, M. (2025, August 17-20). *I2DTM: Immersive and Interactive Digital Twin Metaverse Framework* [Conference Presentation]. International Design Engineering Technical Conference & Computers and Information in Engineering Conference 2025, Anaheim, CA, United States.
- **Lee, H.**, Sim, Y., Han, C., Lee, J., & Jun, M.^{‡*} (2025, July 6-11). *ARMPASS: Autonomous Robot Motion Planning with Adaptive Safe Set* [Conference Presentation]. International Conference on Precision Engineering and Sustainable Manufacturing 2025, Chiang Mai, Thailand.
- Sim, Y., Kim, E., Lee, J., **Lee, H.**, Yun, D., Myeong, N., Hwang, Y., Park, H., & Jun, M. B. G. (2025, June). Sound Data Augmentation Using Frequency Response Superposition for Machine Tool Component Running State Recognition. In *International Manufacturing Science and Engineering Conference* (Vol. 89022, p. V002T16A006). American Society of Mechanical Engineers.
- **Lee, H.**, Choi, Y. W., Lugo, E., Lee, J., Lee, S. W., & Jun, M. B. G. (2025, June). GMGP: Generalized Model for Grasp Planning of Vacuum and Parallel Jaw Grippers. In *International Manufacturing Science and Engineering Conference* (Vol. 89022, p. V002T16A008). American Society of Mechanical Engineers.
- Han, C.[‡], Gabor, T., **Lee, H.**, Lee, J., Akin, S., & Jun, M.* (2025, June 1-4). *Pulsed Cold Spray System for Physical Unclonable Function Generation* [Paper Presentation]. CIRP International Symposium on Electro Physical and Chemical Machining 2025, Vancouver, BC, Canada.
- Jun, M.^{‡*}, Lee, J., Kim, E., Sim, Y., Han, C., & **Lee, H.** (2024, July 7-12). *Generalized and Generative AI for Smart Manufacturing* [Keynote speech] International Conference on Precision Engineering and Sustainable Manufacturing 2024, Busan, Republic of Korea.
- Han, C., **Lee, H.**, Lee, J., Jun, M. B. G., & Yun, H. (2024, June). A Study on Digital Twin for Autonomous Robotic Chip Removal Deep Learning Framework on CNC Machine. In *International Manufacturing Science and Engineering Conference* (Vol. 88100, p. V001T02A003). American Society of Mechanical Engineers.
- **Lee, H.**^{‡*}, Toner, T., Tilbury, D., & Barton, K. (2022, October 2-5). *Multi-sensor aided deep pose tracking* [Paper Presentation]. Modeling, Estimation and Control Conference 2022, Jersey City, NJ, United States.

POSTER SESSIONS

- Han, C.[‡], Sim, Y., Lee, J., **Lee, H.**, & Jun, M. (2025, August 5-9). Industrial Audio Foundation for Agile Machine Listening [Poster session]. US-Korea Conference 2025, Atlanta, GA, United States.
- Sim, Y., Lee, J., **Lee, H.**, Yun, D., Myeong, N., Hwang, Y., Park, H., Jun, M., & Kim, E. (2024, December 2). *Sound Data Augmentation Using Frequency Response Superposition for Machine Tool State Recognition* [Poster session]. Manufacturing And Materials Research Laboratories Symposium 2024, West Lafayette, IN, United States.
- Lee, J., Jeon, J., Sim, Y., **Lee, H.**, Han, C., Yun, D., & Jun, M. (2024, October 15-16). *Generative AI Based Real-Time Human-Data Interaction for Smart Manufacturing* [Poster session]. Digital Twin for Manufacturing Sustainability, Safety, and Resilience 2024, West Lafayette, IN, United States.
- **Lee, H.**, Sim, Y., Lee, J., Han, C., & Jun, M. (2024, October 15-16). *Digital Twin & Vision Guided Autonomous Robotic Path Planning* [Poster session]. Digital Twin for Manufacturing Sustainability, Safety, and Resilience 2024, West Lafayette, IN, United States.
- Lee, J., Kim, E., Han, C., **Lee, H.**, & Sim, Y. (2024, March 24-26). *Generalized and Generative Smart Manufacturing* [Poster session]. North Academy of Engineering Regional Meeting 2024, West Lafayette, IN, United States.
- Han, C., **Lee, H.**, & Jun, M. (2024, May 24-26). *Robotics in Smart Manufacturing* [Poster session]. Congressional Artificial Intelligence Caucus 2024, Washington, D.C., United States.

FELLOWSHIPS

- **Winkelman Fellowship** 2023 - 2024
- **Multidisciplinary Design Program (MDP) Summer Fellowship** 2021 - 2021

OTHER HONORS & AWARDS

- UKC 2025 Best Poster Award N/A
- Italian Technology Award 2025 N/A
- NSF Travel Award for NAMRC/MSEC 2025 N/A
- DigiTwin 2024 for Manufacturing, Sustainability, Safety, and Resilience N/A
- Student Best Poster Award
- Young Professional NET Award 2024 - 2025
- MECC 2023 Travel Grant N/A

PATENTS

- Jun, M., Lee, J., **Lee, H.**, Akin, S., Han, C., Gabor, T., & Sim, Y. (2025, September 18). *Cold spray-enabled physically unclonable identifier and its spectral authentication via implicit neural representation* (U.S. Patent Application No. [Number Assigned]). U.S. Patent and Trademark Office.

INDUSTRIAL PROJECTS

- **BCI Solution, Inc** 2024 - 2025
 - Developed a digital twin of FANUC CRX-10iA/L using MuJoCo physics simulator and Python.
 - Established bi-directional communication among FANUC CRX-10iA, its digital twin, and CESMII platform for automation and verification of core-making operations.
- **Korea Institute of Machinery and Materials (KIMM)** 2020 - 2022

- Developed a generalized sampling algorithm to extend a cutting edge object segmentation AI to 6-DOF object pose estimation method for machine tending.
- Developed a hybrid computer vision pipeline to localize metal debris inside CNC machine that must be removed.
- **BCI Solution, Inc** 2014 - 2020
 - Built a server-client system using Python and KAREL to communicate FANUC controller via external PC.
- **Terra Drive Systems (TDS)** 2023 - 2023
 - Installation of stethoscope-based sound sensor and mini PC to establish a data pipeline.
 - Developed a sound recognition AI model to monitor status of Makino a81 CNC machine.
 - Generated a web-based dashboard for intuitive data analysis and visualization.

PROFESSIONAL SOCIETY MEMBERSHIPS & SERVICES

- IEEE Transactions on Automation Science and Engineering / Reviewer 2025 - Present
- Graduate Association for Manufacturing Excellence / Treasurer 2025 - Present
- Korean-American Scientists and Engineers Association (KSEA) / Member 2024 - Present

LEADERSHIP & MENTORSHIP

- Student Mentor 2024 - Present
 - Richard Li - B.S. in Computer Science (Summer 2024), Evin Lugo - B.S.E. in Mechanical Engineering (Fall 2024 - Spring 2025), Yoonseo Lee - B.S.E. in Mechanical Engineering (Spring 2025 - Summer 2025), Junho Sohn - M.S. in Mechanical Engineering (Fall 2025 - present)
- Sub-Team Leader (Barton Research Group) 2021 - 2022
 - Facilitated communication among 5 sub-team members and Ph.D. mentors.
 - Led weekly team meetings for discussion among members and provided advice regarding their projects.

CERTIFICATIONS

- Deep Learning Specialization, DeepLearning.AI
- Modern Robotics, Northwestern University

SKILLS

- **Programming:** C/C++ (Beginner), Python (Professional), MATLAB (Professional)
- **Frameworks/Libraries:** ROS, PyTorch, TensorFlow
- **Languages:** English (bilingual), Korean (native), Japanese (conversational)